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Rig-Less Surface Jet Pump Deployment: A Sustainable and Cost-Effective Solution for Well Revitalization: A Case study from UAE

Saiq Shakeel Abbasi, Zubair Khan, and Abdul Qudoos Awan, Weatherford Bin Hamoodah, Abu Dhabi, UAE.

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Abstract

Once a Well's reservoir pressure depletes, it loses its natural production and requires a workover operation which is time-consuming, costly and does not guarantee production revival. Exploration and Production companies seek a compatible, sustainable, and cost-effective artificial lift solution to restores the Wells' production to increase the upstream activities. This paper highlights the quantifiable impacts, implementation strategies, and lessons learned, demonstrating how the Surface Jet Pump system overcomes flowline back pressures.

The Surface Jet Pump (SJP) works on the Venturi principle, where drop in pressure increases the velocity of the fluid, creating a drawdown that allows the Well fluid to enter the Jet Pump. It is connected to the Well, a power fluid line, and a discharge line and is positioned near the Wellhead. SJP requires a high-pressure power fluid to create the required drawdown and overcome the back pressure from the flowlines to offload the Well. Typically, this high-pressure power fluid is produced by a surface pumping unit, which requires an emissive energy source, especially in remote operations.

This study presents a trial application of SJP on Well A which was out of production due to high back pressure from the flowlines and required frequent interventions (Nitrogen kick offs). Based on the data, simulations were completed for the optimized nozzle and throat combination, ensuring the required draw down and discharge pressure to revive the production from the well. In this application instead of using the surface pumping unit, a high-pressure water source was identified from the operator's water injection system and was used as power fluid. By utilizing this fluid, the need for an emissive source to run the surface pumping unit was eliminated, reducing CO_2 equivalent emissions by 28.42 metric tons per month. This rig-less SJP system restored continuous production, achieving a production rate of over 1,200 BOPD. This increased revenue, reduced OPEX cost over $ 50000 per month for the operator. This innovative approach introduced low-noise, low-carbon footprint technology, improving well performance and operational efficiency without requiring major infrastructure modifications. In 2024, this project resulted in nearly 300 metric tons of CO_2e reductions for the operator. It delivers low-cost, high-efficiency production enhancement with minimal environmental footprint.

The rig-less installation of the compact SJP skid saved valuable downtime, allowing for quicker deployment making the system both cost-effective by reducing interventions and environmentally

sustainable, while eliminating the need for traditional surface pumps. Additionally, the SJP, with no moving parts, contributed to safer equipment and low maintenance requirements. This deployment aligned with the "Net-Zero Emissions by 2050 Sustainability Strategy", aimed at reducing carbon emissions and addressing climate change.

Keywords: Surface Jet Pump, Sustainable, Net-Zero emissions, Venturi Principle

Introduction

Artificial lifts are used in about 85% of wells globally, making their role in production efficiency and profitability (Espin et al. 1994). Hydraulic Jet Pump lift systems are a type of artificial lift widely applied in the petroleum industry. A jet pump is a Rigless intervention technique that creates drawdown and offloads the well when reservoir pressures are depleted and wells no longer flow naturally. Jet pumps have been used since the early 1970s and are considered one of the most cost-effective artificial lift methods (Kumar et al. 2015; Society of Petroleum Engineers n.d.). Their simplicity, lack of moving parts, and adaptability to various well conditions make them particularly suitable for challenging environments. For instance, jet pumps can effectively deploy in high deviated wells up to 90° of deviation and 24°/100 ft dogleg severity and handle high gas-to-liquid ratios better than other artificial lift solutions. They are also capable of functioning at extreme temperatures up to 500°F by using high-temperature elastomers (Louis, 2021). It can be run and retrieved via fluid circulation or slickline, making it suitable for deviated wells. The Jet pumps are sand, corrosive fluids, and gas tolerant, and are excellent for producing viscous crude. Jet pumps have an adaptative system and can run with BHAs, sliding sleeves, tubing punches and tubing packers, repairable at the well site, and requires low maintenance with easy repairs. Additionally, it allows continuous treatment of tubing and annulus via power fluid. Jet Pumps are not known for high efficiency, but their simple design make them practical, especially when a high-pressure fluid source is already available. In such cases, they offer a dependable and low-maintenance solution that can be more cost-effective than complex mechanical systems. Jet pump operations are a profitable alternative to increase production across the fields (Guzman et al. 2021).

Exploration and production (E&P) companies in the world are continuously working on reduction of operational emissions and achieving a net-zero Sustainable Development Goals (SDGs). These initiatives include electrification of operations and replacing the diesel-powered generators, introduction of clean energy like solar and wind to meet energy requirements, carbon capture and storage (CCS) technologies are also being implemented to mitigate CO_2 emissions (Akpasi and Isa 2022), flaring reduction (McKinsey & Company 2024) and water management among the top initiatives (ADNOC 2024; Equinor ASA 2024; Walker et al. 2024).

This study shows a test use of Surface Jet Pump on Well A, which was not producing because the flowlines had too much back pressure and needed repeated interventions (Nitrogen kick offs). In this instance, a high-pressure water source from the operator's water injection system was utilized as the power fluid replacing the need of surface pumping unit. Rig-less Surface Jet pump system was assessed and deployed on an inactive well in June 2024. The results of the study were successful, and the well is producing above 1200 BOPD, adding $100K of revenue. This aligns with the client's 2030 Sustainability Strategy to reduce operational emissions intensity by 25%, which encouraged us to propose a Surface Jet Pump application using motive fluid from a nearby water injection source (Weatherford 2024a, 2024b).

Surface Jet Pump

The Surface Jet Pump (SJP) operates based on the Venturi effect, where a reduction in pressure leads to an increase in fluid velocity. This creates a drawdown that enables Well fluids to be drawn into the jet pump. The fundamental theory behind jet pump fluid mechanics was first formalized by Cunningham in the Pump Handbook (Karassik et al. 2001) , where each component of the jet pump is described with its own flow regime and corresponding equations. These equations explain how pressure, velocity, and other fluid properties change throughout the pump. Surface jet pumps have multiple applications in oil and gas industry to improve production efficiency (Peeran et al. 2013; Carpenter 2014).

Jet pumps are simple, yet highly effective devices used in artificial lift systems, particularly in mature oil fields and wells experiencing declining reservoir pressure. This process is governed by fluid mechanics principles, particularly the Venturi effect, which is central to jet pump operation. Moreover, jet pumps are advantageous in terms of maintenance and deployment. Unlike ESPs or rod pumps, jet pumps do not require a rig for retrieval, which significantly reduces operational costs and downtime. Their compact and rig-free design allows for easy transportation and installation, even in remote Well sites. This makes them ideal for marginal fields where infrastructure is limited, and cost-efficiency is paramount. The jet pump is also suitable for wellbore cleanup post-completion and for producing heavy, viscous, or corrosive fluids. By enabling early-stage production without significant capital expenditure, it helps defer capital expenditure (CAPEX) and reduces operational expenses (OPEX) through rig-less installation and servicing.

In summary, jet pumps are a mature yet evolving technology in artificial lift systems. Their ability to operate without moving parts, handle complex flow regimes, and integrate seamlessly with existing infrastructure makes them a preferred choice for many operators. As industry continues to focus on maximizing recovery from aging assets, jet pumps are poised to play an increasingly significant role in production optimization strategies. With ongoing research and field validation, their application is expected to expand, offering reliable and economical solutions for a wide range of production challenges.

Working principle of the Jet Pump

A jet pump includes three static components: the nozzle, throat, and diffuser (Lyons et al. 2016). The operation begins with highly pressurized power fluid typically water or oil being converted into kinetic energy as it passes through the nozzle. This high-velocity stream creates a low-pressure zone at the throat, where reservoir fluid, which is at a much lower pressure, is entrained. The two fluids mix in the throat, forming a homogeneous mixture that enters the diffuser. Where the kinetic energy of the combined fluids is converted back into pressure energy, which is sufficient to lift the fluid to the surface.

The power fluid from surface pump is pumped at a specified rate (Q_N) to the jet pump where it enters the nozzle with pressure (P_N) as shown in Fig.1, at the exit of nozzle it gets converted into high velocity, low pressure flow from low velocity, high pressure flow. The low static pressure, high velocity fluid enables well fluids to be pushed by the reservoir at the required rate in to the well. From the well the reservoir fluid then travels and flows into the annular space between the nozzle and throat with rate (Q_s) and pressure (P_s) as shown in Fig. 2.

The power fluid, along with the fluid from reservoir then enters the throat, where the momentum transfers from power fluid to produced fluid and the intermixing of both fluids takes place inside the throat. The flow from throat through a diffuser section, where it is converted to a high pressure, low velocity state and exists the pump. The high discharge pressure (P_d) must be enough to lift the co-mingled fluid flow rate (Q_t) to the surface (Lea et al. 2008; T. Pugh 2009).

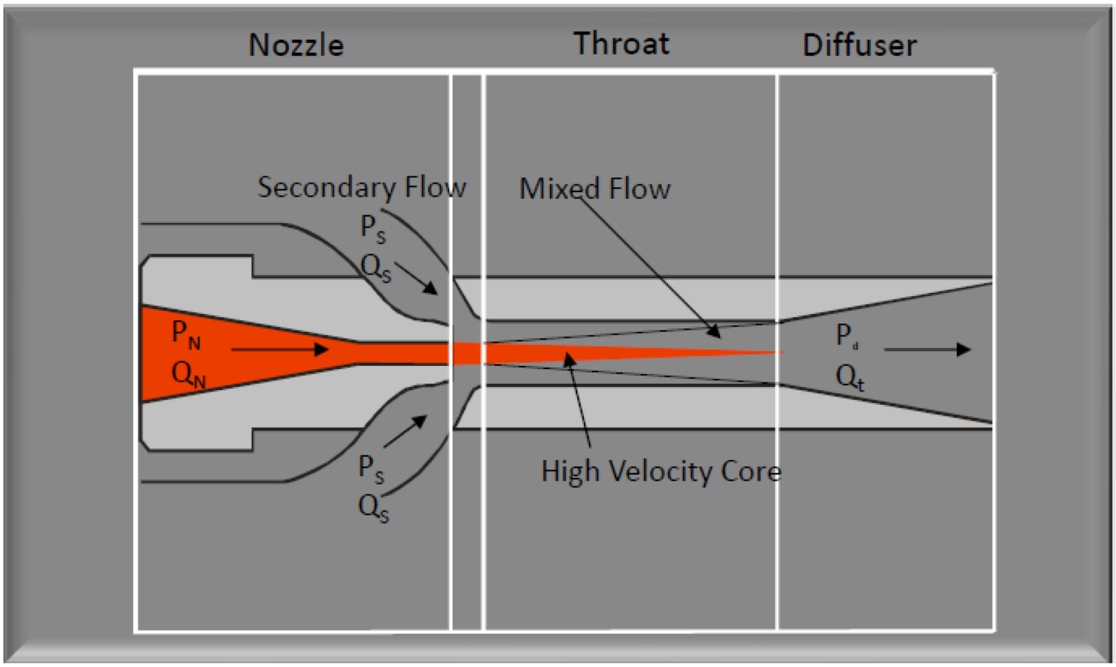

Figure 1—Velocity Exchange in the components of Jet Pump.

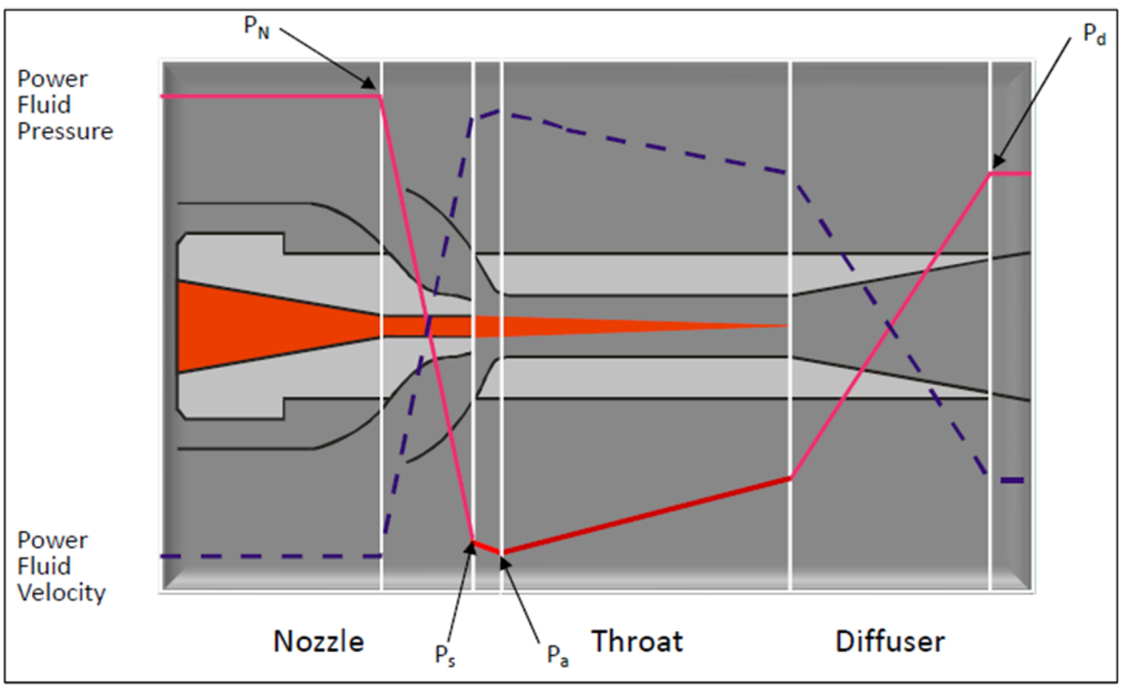

Figure 2—Pressure and velocity changes inside the parts of jet pump Nozzle and Throat.

Design and Simulation

Simulations were created to choose the best nozzle and throat configurations to maximize jet pump performance. Key variables like the Well's Inflow Performance Relationship (IPR), fluid properties, injection pressure, and power fluid characteristics were all included in the analysis Fig.3. To assess the effects of these characteristics and guarantee the best possible design and operational efficiency, a thorough sensitivity analysis was conducted for the candidate wells. The evaluation and revalidation phase to select the optimal nozzle and throat combination for the Surface Jet pump based on data provided by Client. The equipment and material are carefully chosen and designed to meet the client's standards, ratings, regulations and working environment. The performance summary of the jet pumps at different injection pressures for the well is given below in Table 1.

```
      Customer     : XYZ                                       Date  : 08/04/2024
      Field & Well : Well A                                    Run By: ABC
      Location     : X                                         Run ID: 123

 1. Perforation Depth        (ft) :     N/A   13.Producing GOR  (scf/STB) :  658.0
 2. Pump Vertical Depth      (ft) :     N/A   14.Gas Sp. Gravity (air=1.) :  0.800
 3. Pump Instl  (1) Standard Flow             15.Separator Press   (psig) :  630.0
    (2)Rev. Flow (3)Parallel Flow:  2         16.Well Static BHP   (psig) :  N/A
 4. Casing ID                (in) :  N/A      17.Well Head Pressure(psig) :  730.0
 5. Tubing OD                (in) :  3.500    18.Well Test Flow Rate(STB/d): 1400.0
 6. Tubing ID                (in) :  3.000    19.Well Head Temp  (deg. F) :  125.0
 7. Return Tubing ID         (in) :  N/A      20.Bottom Hole Temp(deg. F) :  N/A
 8. Tubing Length            (ft) :  1000.0   21.Bypass Gas? (1)yes (2)no :  2
 9. Pipe Cond (1)new(2)avg(3)old :  3        22.Power Fld (1)oil (2)water:  2
10.Oil Gravity              (API) : 42.140    23.Power Fld API/Sp. Gravity:  1.150
11.Water Cut                  (%) :  42.00    24.Bubble Point Press(psia) : 2523.2
12.Water Specific Gravity         :  1.150    25.Flowline Press   (psig) :  800.0
================================================================================
 12D  Jet Pump Performance

 Sensitivity Based on the Four Specified Surface Injection Pressures:

 Injection Pressure   =   1500. psig
          Production Rate        =   2010. STB/d
          Injection Rate         =   1300. bpd
          Horsepower to Jet Pump =     38. hp
          Pump Intake Pressure   =    699. psig
          Discharge Pressure     =    806. psig
          Cavitation Rate        =   2856. STB/d

 Injection Pressure   =   1600. psig
          Production Rate        =   1914. STB/d
          Injection Rate         =   1374. bpd
          Horsepower to Jet Pump =     43. hp
          Pump Intake Pressure   =    703. psig
          Discharge Pressure     =    807. psig
          Cavitation Rate        =   2873. STB/d

 Injection Pressure   =   1700. psig
          Production Rate        =   2520. STB/d
          Injection Rate         =   1472. bpd
          Horsepower to Jet Pump =     49. hp
          Pump Intake Pressure   =    672. psig
          Discharge Pressure     =    807. psig
          Cavitation Rate        =   2765. STB/d

 Injection Pressure   =   1800. psig
          Production Rate        =   2440. STB/d
          Injection Rate         =   1538. bpd
          Horsepower to Jet Pump =     54. hp
          Pump Intake Pressure   =    676. psig
          Discharge Pressure     =    808. psig
          Cavitation Rate        =   2779. STB/d
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Figure 3—Jet Pump Simulations using data provided by the client.

Table 1—Summary of the Surface Jet Pump Simulations with Production, Injection pressure, cavitation, and horsepower requirement details.

Injection Pressure [psig]	Production Rate [STB/d]	Injection Rate [bpd]	Horsepower [hp]	Pump Intake Pressure [psig]	Discharge Pressure [psig]	Cavitation Rate [STB/d]
1500	2010	1300	38	699	806	2856
1600	1914	1374	43	703	807	2873
1700	2520	1472	49	672	807	2765
1800	2440	1538	54	676	808	2779

The candidate Well A had high flowline back pressure of 800 psig, with a Well Head pressure of 730 psig it could not flow naturally. The client water injection network had a maximum injection pressure of 1800 psig to be provided along with 1600 bpd of water rate. These parameters were essential in determining the size of the Nozzle and Throat to create a certain drawdown and discharge pressure for the surface jet pump to overcome the high flowline backpressure.

The best production rate is achieved at 1700 psig injection pressure (2520 STB/d). Cavitation rate decreases slightly with increasing injection pressure but remains high across all cases. Pump intake pressure drops slightly with higher injection pressure, which may affect pump efficiency. Horsepower requirement increases with injection pressure, indicating higher energy demand.

A critical observation from Table 1 is the discharge pressure. This pressure must exceed the flowline pressure for the SJP to effectively overcome backpressure and sustain flow. As in this case, the flowline pressure was 800 psig, while the SJP achieved a discharge pressure of 807 psig, provided necessary pressure differential to compel the production through the flowline.

Surface Jet Pump System Configuration

The hydraulic Surface Jet Pump lift system comprises a Power unit (PU), Vessel unit (VU), and a jet pump. The PU and VU are designed as standalone systems. The PU boosts pressure and displaces power fluid to the surface jet pump. The VU receives both produced and power fluid returns from the well, allowing separation of the power fluid from production so it can be reused. Upon returning from the well, the fluid mixture enters a horizontal reservoir vessel. It separates gas and produces fluids, and excess fluid is discharged through the different outlets depending upon the production, in usual cases the vessel have separate vents for oil, gas and water. This fluid is then directed to the flow line; the output of the vessel may include a no return valve maintain a system pressure greater than that of the flow line. The power fluid is directed back to the surface pump into the SJP.

Fig. 4 shows such configuration where a multiplex pump is used to pressurize the power fluid and channeled into the Surface Jet pump to create a drawdown for the Well fluids to enter the SJP and mixes with the power fluid and directed towards the flowline at a certain pressure to overcome the flowline pressure. This configuration uses a multiplex pump to attain a certain discharge pressure required for the drawdown; this pump is powered by a prime mover which requires a power source, and for remote applications like in present study needed a diesel-powered generator.

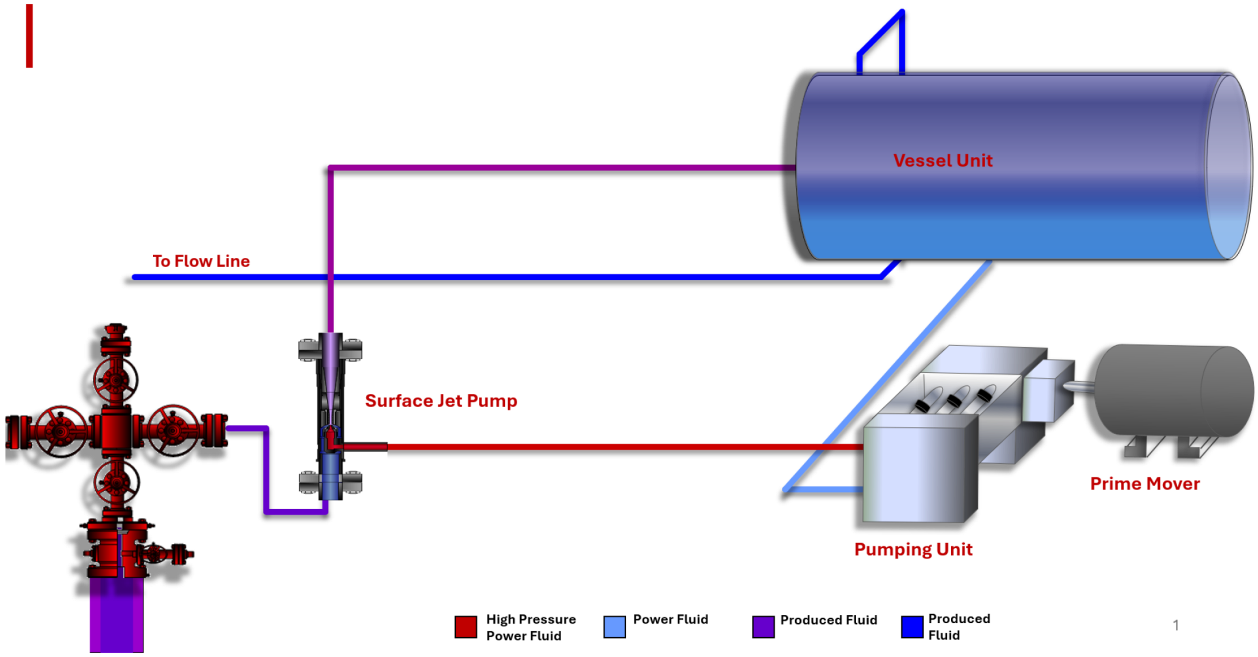

Figure 4—Surface layout for Hydraulic Surface Jet Pump Operations with a surface pumping unit and prime mover.

In an alternative approach used in this study as shown in Fig. 5, we have utilized the existing water injection system of the client. With a network of Plate wells which are a group of wells with water injectors and a single producer in close vicinity to each other. This is an Enhanced Oil recovery technique where the well reservoir pressure is maintained through a network of wells in which three to four wells function as water injectors while one is an Oil producer.

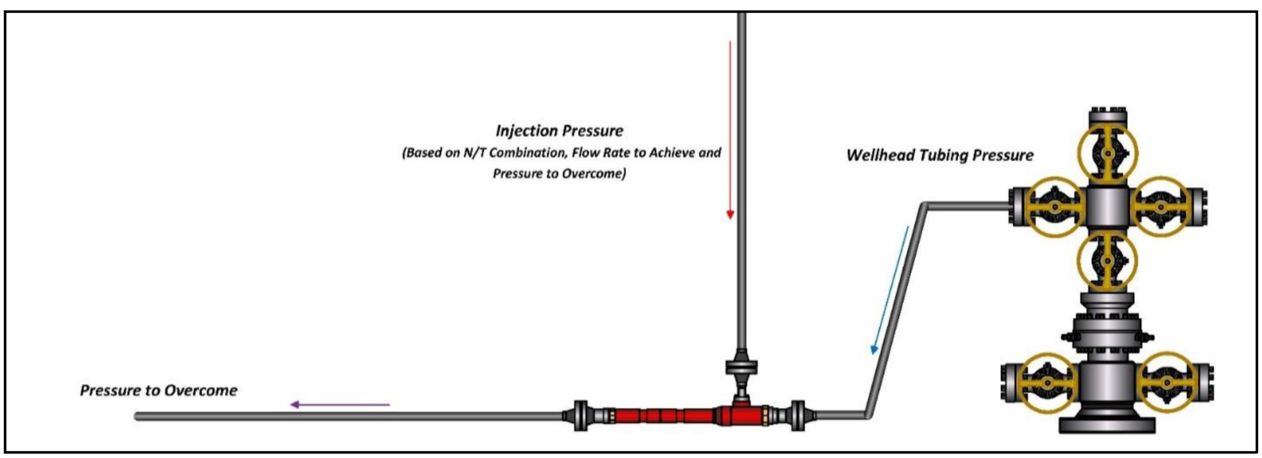

Figure 5—Surface layout for operation of Hydraulic Surface Jet Pump Operations using Water injection lines from the Clients water injection network.

To kickstart the surface jet pump operation, the water injection lines are connected with the Jet pump in the along with connecting it with Wellhead and flowlines. All the lineup is checked, the Well was kick started using Coil tubing nitrogen kick off prior to deviating the lineup to the surface jet pump with the power fluid from the water injection system, and pressures at the injection point, wellhead, and flow line will be closely monitored. Returns from the jet pump were directed to the oil producer's flow line, with continuous follow-up until operational stability is achieved.

Results

The candidate well, Well A, an oil producer, was not able to produce due to high back pressure from the flow lines after injection of water. The client relied on multiple techniques to offload the well, with frequent nitrogen kickoffs, to restore production. This solution was costly, time consuming, and was impacting on the environment caused by nitrogen pumping units, which require continuous energy sources and produce emissions. This water injection system uses centrifugal pumps to pressurize the water to inject into these wells.

Well A is paired with the neighboring water injection PLAT-X, which functions as a high-pressure water source. This pressurized injection water was used as a power fluid for the surface jet pump. The primary objective is to extend the pipeline from PLAT-X to Well A. The water line originating from PLAT-X were directed into a surface jet pump to provide the necessary driving force for Well A. As the power fluid moves through the nozzle, the Venturi effect created a pressure drop at the pump throat, reaching it below the wellhead pressure. This induced flow from well into the SJP, the combined mixture progresses into the diffuser of the surface jet pump, generating discharge pressure sufficient to surpass the flowline pressure. This strategic implementation aims to overcome the flow line pressure, thereby enhancing production levels, especially in Well characterized by high backpressure.

The Surface Jet Pump technology enabled the well to produce 1,200 BOPD, significantly boosting the client's production. By eliminating the need for frequent Nitrogen Kick Offs and traditional multiphase surface pumps, the client achieved substantial operational costs and reduced downtime. The rig-less installation of the Surface Jet Pump saved valuable Rig-time, allowing for quicker deployment and reduced intervention frequency. The use of motive fluid as a high-pressure source provided a sustainable and environmentally friendly solution, aligning with the client's long-term goals (Weatherford 2024b).

Discussion

A transformative approach to overcome high flowline backpressure without relying on the conventional surface pumping units as shown in Fig 4 has been demonstrated in this deployment of Surface Jet Pump system (SJP) on Well A. The jet pump intake was connected to the well head production string, and its discharge was tied with the flowline, utilizing pressurized water from the PLAT X as power fluid. Engineering estimates indicated that using a traditional surface setup would have required an additional 400 man-hours for site preparation and commissioning. This would have resulted in operational downtime, increased space requirements, and a continuous diesel supply of approximately 350 liters per day throughout the operation.

This system was implemented without any major changes in the infrastructure or intervention resulting in lower operational costs and reduced downtime. The mobile and compact SJP skid is easy to install and relocated to various locations once the desired results are achieved. The working principle in this application have enabled client in utilizing pressurized water injection to run the system without using traditional multiphase pumps reducing approximately 28.42 metric tons of CO_2e emissions per month.

Conclusion and Recommendations

Well-A, a single string oil producer, became inactive due to high back pressure from the flowline. The deployment of Surface Jet Pump (SJP) technology restored production enabling 1,200 BOPD production. The rig-less installation of the SJP saved valuable rig-time, allowing faster deployment and reduced intervention frequency. This innovative method improved Well performance and operational efficiency without requiring significant infrastructure changes, and provided sustainable, low-noise, and low-carbon footprint technology. This initiative helped the client to reduce CO_2e emissions by approximately 300 metric tons in 2024 (Weatherford 2024b), while leaving a small environmental impact due to the reduction in diesel fuel usage, supporting sustainability goals.

The Surface Jet Pump offers a cost-effective, rig-less solution for enhancing production across a wide range of applications. With no moving parts and compactness, it simplifies operations and minimizes maintenance requirements. This technology is particularly effective in overcoming high flowline pressure, enhancing natural well pressure, and restoring production.

Abbreviations

ADNOC	Abu Dhabi National Oil Company
API	American Petroleum Institute
BHA	Bottomhole Assembly
BOPD	Barrels of Oil Per Day
Bpd	Barrels per Day
CAPEX	Capital Expenditure
CCS	Carbon Capture and Storage
CO_2	Carbon Dioxide
CO_2e	Carbon Dioxide Equivalent
ESP	Electric Submersible Pump
E&P	Exploration and Production
hp	Horsepower
IPR	Inflow Performance Relationship
OPEX	Operational Expenditure
PLAT-X	Plate Injection Well (contextually inferred as part of a water injection network)
PN	Nozzle Pressure
PU	Power Unit
psig	Pounds per Square Inch Gauge
QN	Nozzle Flow Rate
SJP	Surface Jet Pump
STB	Stock Tank Barrel
STB/d	Stock Tank Barrels per Day
VU	Vessel Unit

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